EFFICIENT GRAPH-BASED IMAGE SEGMENTATION VIA SPEEDED-UP TURBO PIXELS

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ABSTRACT

An efficient graph based image segmentation algorithm exploiting a novel and fast turbo pixel extraction method is introduced. The images are modeled as weighted graphs whose nodes correspond to super pixels; and normalized cuts are utilized to obtain final segmentation. Utilizing super pixels provides an efficient and compact representation; the graph complexity decreases by hundreds in terms of node number. Connected K-means with convexity constraint is the key tool for the proposed super pixel extraction. Once the pixels are grouped into super pixels, iterative bi-partitioning of the weighted graph, as introduced in normalized cuts, is performed to obtain segmentation map. Supported by various experiments, the proposed two stage segmentation scheme can be considered to be one of the most efficient graph based segmentation algorithms providing high quality results.

Index Terms— Super pixel, normalized cuts, color segmentation

1. INTRODUCTION

Image segmentation which provides extraction of semantically meaningful objects automatically from an image, has been one of the most popular problems in computer vision. There have been many different approaches for the solution of segmentation and among them graph-based methods have been popular last decade due to their applicability on global optimization [1]-[4].

Graph-based methods introduce a top-to-bottom approach, which models images as weighted graphs and provides segmentation by recursively partitioning the graph into sub-graphs. Top-to-bottom approach is an important mechanism as it is also observed in human perception, interpreting a scene from whole to detail. This assumption, however, results in details to be lost in the segmented objects, especially for textured surfaces. Besides, the computational complexity increases tremendously as the image sizes are increased.

In order to preserve details and decrease computational complexity, some methods have been proposed by [5][6] that unifies the speed and detail preserving advantages of bottom-to-top methods [7] and the semantically meaningful partitioning of graph-based methods. In [5], the graph is constructed via super pixels [8] which encapsulates similarly colored pixels and represent the image as a group of patches. Such an approach decreases the number of nodes in a graph abruptly from millions to thousands; hence, the computational complexity. Moreover, the super pixels capture the details among pixels and prevent local variations to be lost in huge graphs. However, such an approach might also cause irregularities in the graph representation resulting in biases in the segmentation, as clearly explained in [5].

There are various studies on the extraction of super pixels, i.e. representing images with limited number of pixels groups [8][9]. In a recent work [8], a robust method is proposed yielding convex and quasi-uniform super pixels which are important for obtaining nearly regular graphs. In addition, quasi-uniform super pixels has an important role in preventing under-segmentation errors during graph bi-partitioning originating from tiny and irregular shaped super pixels [8]. The *TurboPixel* algorithm [8] can be accepted to be the superior over-segmentation algorithm suitable for image representation in terms of speed, compactness of segments and over-segmentation errors according to the results given in [8].

In this work, previous study [5] is improved by modifying the super pixel extraction step with a much faster method compared to [8] and [7], providing quasi-uniform over-segments with a similar performance of *TurboPixel*. The proposed method exploits a connected K-means algorithm with an additional convexity constraint; thus, images are modeled via more regular graphs providing more stable *N-cut* segmentation over super pixels.

The details of the proposed super pixel extraction algorithm are given with comparative analysis in Section 2. After the overall image segmentation algorithm is presented in Section 3, experimental results are given in Section 4. The final section is devoted to the concluding remarks.

2. SUPER PIXEL EXTRACTION

Representing images via limited number of pixel groups rather than pixels has been a common approach in stereo estimation problem [12]. Such an approach preserves discontinuities among object boundaries, and un-textured regions (which are problematic in stereo) are handled easily. Hence, the same idea can be exploited in image segmentation to engage local and global methods [5][6], by gathering the advantages of both. The local part involves grouping similarly colored pixels via merging; the global part as in most graph based segmentation routines involves splitting the graph whose nodes corresponds to pixel groups.

As mentioned in [5] and [8], the over-segments which are distributed non-uniformly among an image cause the model graph to be irregular. Hence, under-segmentation errors due to biases over dense (small sized over-segments) super pixel regions can be observed. In [5], a neighbor number limiting method has been proposed, while the nodes are linked to each other in order to regularize the graph. This method handles such cases; however, it does not remove irregularity completely. In addition, local methods, such as Mean Shift [7], are time consuming techniques that provide initial over-segmentation. Though, faster methods are required to decrease computational complexity of algorithms exploiting over-segments, such as disparity estimation, segmentation and image modeling.

Regarding the need for quasi-uniformity of super pixels and fast operation time, a connected K-means algorithm with convexity constraint is developed in this study. Initially, an image is divided into rectangular regions according to the desired number of pixels. Hence, each super pixel has rectangular shape with centers equally spaced among the image, Figure 1. This step is considered as the initialization of super pixel extraction; in the next step, the pixels at the over segment boundaries are assigned to the new segments by minimizing a cost function as given in (1).

$$C_{x,y}(i) = \lambda_1.\left|I(x,y) - I_i\right| + \lambda_2.\left|(x - C_x^i)^2 + (y - C_y^i)^2\right| \quad (1)$$

In (1), I_i indicates to the mean intensity of the i^{th} segment, x and y are the locations of the pixel tested among different segments. C_x^i and C_y^i are the center locations of the i^{th} segment; where λ corresponds to the weighting of intensity similarity and convexity constraints. The first term in (1) ensures similarly colored pixels to be merged, whereas the second term provides super pixels to have convex shapes by preventing distant pixels to be merged. The convexity of segments can be arranged by changing λ_2.

During the updates of super pixels, individual pixels located at the segment boundaries are tested, which assures super pixels to be connected, as given in Figure 1. Such an approach, though, provides the algorithm to be fast, since limited number of pixels is tested among neighboring segments (max number of 4 neighbors). Once all the boundary pixels are tested, super pixel mean intensity and center locations, which are updated as new pixels, are merged and removed. When the number of interchanged pixels is below a threshold, the iterations stop and the super pixel generation step is finalized. An example for discrete steps from initialization to final super pixels is illustrated in Figure 1.

The proposed method is compared to the state of the art super pixel extraction technique [8] in terms of operational speed and segmentation quality. The test images are taken from Berkeley Segmentation Database [13] (with size of 481x321) and [8] (with size of 2560x1920). The over-segmentation results are illustrated in Figure 2 and Figure 3. According to the visual comparison, it is clear that the proposed method has same and even better performance at some object boundaries compared to [8].

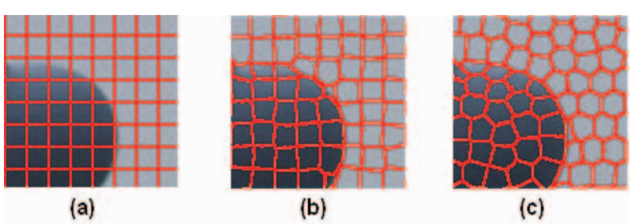

Figure 1: Pixels at segment boundaries (red colored) are updated at each turn; (a) initialization, (b) an intermediate step, (c) final step.

Compared to [8], the experiments are performed on a *core duo 1.80 GHz 2G Ram* computer, which is at most two times slower than the environment used in [8]. The operation time of the two algorithms are given depending on the image size as in Figure 4, where image size is considered as the fraction of the area of full image size 481x321; when compared to the operation time of *TurboPixels*, the difference is quite obvious. Even for high definition images such as resolution of (2560x1920), the proposed algorithm extracts super pixels within 17 sec. which is half the time of *TurboPixels* spending for (481x321) sized images. The operation time actually is affected minimally by the number of super pixels with the proposed algorithm as illustrated in Figure 4. During the experiments, the number of super pixels is fixed to 1000 for images with sizes (481x321) and 10000 for (2560x1920).

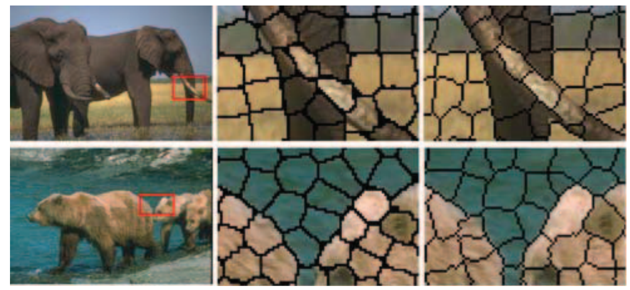

Figure 2: Two images of size (481x321), second column for proposed method, the last column for *TurboPixels*.

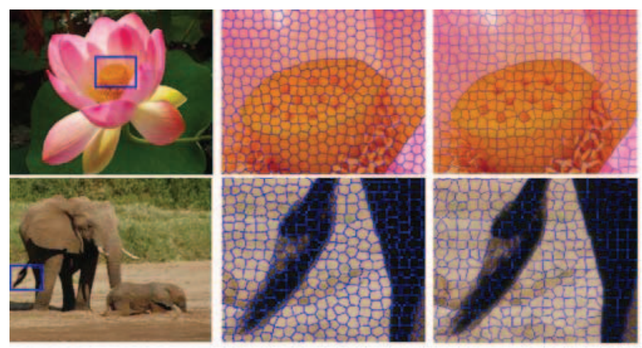

Figure 3: Images of size (2560x1920), second column corresponds to the proposed method, last column is for *TurboPixels*.

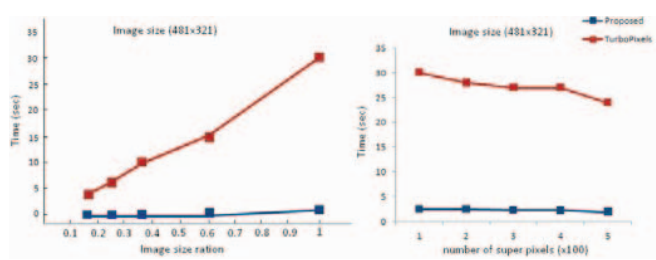

Figure 4: Left: operation times for both methods with respect to image size; Right: time dependency on the number of super pixels.

According to the comparison between *TurboPixels* and the proposed method, it could be concluded that similar super pixel generation is obtained with a significant decrease (almost %98) in computation time by the proposed speeded-up turbo pixels algorithm. Hence, a fifty times faster turbo pixels generation is provided compared to [8] with similar performance.

3. REGION BASED NORMALIZED CUT SEGMENTATION

The top to bottom approach for image segmentation is achieved by mapping the image into a graph and performing iterative partitioning. In that manner, over segments provide more compact image representation compared to original pixel wise models. As proposed in [5], graph modeling can be achieved by using super pixels instead of pixels in order to decrease number of nodes and preserve details. Such an approach can be considered to be an efficient unification of top to bottom and bottom to top methods, exploiting the advantages of both.

The possible problems that can be encountered during such an approach are explained in detail in [5]; such as the irregular distribution of nodes among the graph. Thus, the graph should be as uniform as possible for best performance. The proposed algorithm introduced in the previous section is devoted for convex and quasi-uniform super pixels; hence a good candidate for the solution of graph irregularity. In that manner, once the super pixels are extracted, a graph is constructed whose nodes correspond to the centers of the super pixels. Then, the patches are linked to each other with a weight function based on intensity, the mean intensity within segments, as given in (2). Where $W_{i,j}$ is the link weight between i^{th} and j^{th} segments, and X_i is the mean location of the i^{th} segment. The segments which are close within distance R are linked to each other.

$$W_{i,j} = \begin{cases} \exp(-\left\|I_i - I_j\right\|^2 / \sigma) & \left\|X_i - X_j\right\| < R \\ 0 & elsewhere \end{cases} \quad (2)$$

Then, the normalized cut value [1], between two sub-graphs A and B, is minimized by a solution of the corresponding generalized eigenvalue system [1]. The eigenvalue decomposition provides a partitioning on the corresponding graph. As introduced in [1], this criterion is utilized recursively to split weakest links between sub-graphs in order to perform the segmentation.

4. EXPERIMENTS

In order to establish the robustness and the computational efficiency of the proposed segmentation scheme, comparative tests are performed against state-of-the-art segmentation algorithms [9]-[11]. In that manner, experiments are performed in two categories; in the first part the proposed algorithm is compared to the well known graph based techniques [1] and [4]. In the second part, comparison is achieved in a wider perspective by the inclusion of state of the art segmentation methods. The experiments are evaluated in terms of the operational speed and the visual consistency of the segmentations.

4.1. Graph Based Methods

Normalized cuts (NC) [1] and its multi-resolution extension [4] are compared with the proposed method on samples from Berkeley Segmentation Database [13]. The results are illustrated in Figure 5; and it is clear that the proposed method preserves details and provides more logical semantic object extraction.

The operation times for the three algorithms are given in the table 1 as well. For NC, due to memory problems images are downscaled by factor 3, thus for actual image sizes operation time should be much larger than the given results. For sake of completeness, computation times for down sampled images are also provided (on the right).

4.2. State of the Art Methods

The proposed method is also compared to state of the art segmentation algorithms as well. For that purpose, the experiments conducted in [10] are extended by [9] and [11]. For the images in Figure 6, segmentation results are given in Figure 7. In addition, the average computation times for the algorithms are also given in Table 2. According to the results, the proposed method can perform superior segmentation on scenes involving one to three objects and a background like *starfish*. In other cases, when the scenes become more complex; some details might be lost. In terms of the operation speed, Table 2 might not give the exact comparison, since different platforms (C, MATLAB) affect the speed; but certainly some rough idea. For the proposed method, super pixel extraction is implemented in C and normalized cut is implemented in MATLAB. Considering the results in Table 1 and Table 2, it can be concluded that the proposed algorithm is quite fast, even the fastest graph based algorithm.

Figure 5: The first column is the original images, then consequently proposed method, multi-scale Ncut [4] and Ncut [1].

Figure 6: Three images utilized for the experiments.

Table 1: The operation times of the graph based methods.

Time (sec) Full / Down sampled	***Church***	***Starfish***	***Flower***
N. Cut [1]	NA / 7,1	NA / 7,1	NA / 6,5
Proposed	**3,3 / 0,9**	**3,5 / 0,9**	**3,7 / 1,5**
M.S N. Cut [4]	28,4 / 3,3	28,8 / 3,7	37,4 / 3,8

Table 2: The operation time and environment

	[11]	**[9]**	**[10]**	**Proposed**
Time(sec)	16,2	7,0	11,1	3,66
Env.	C	C	MATLAB	C+MATLAB

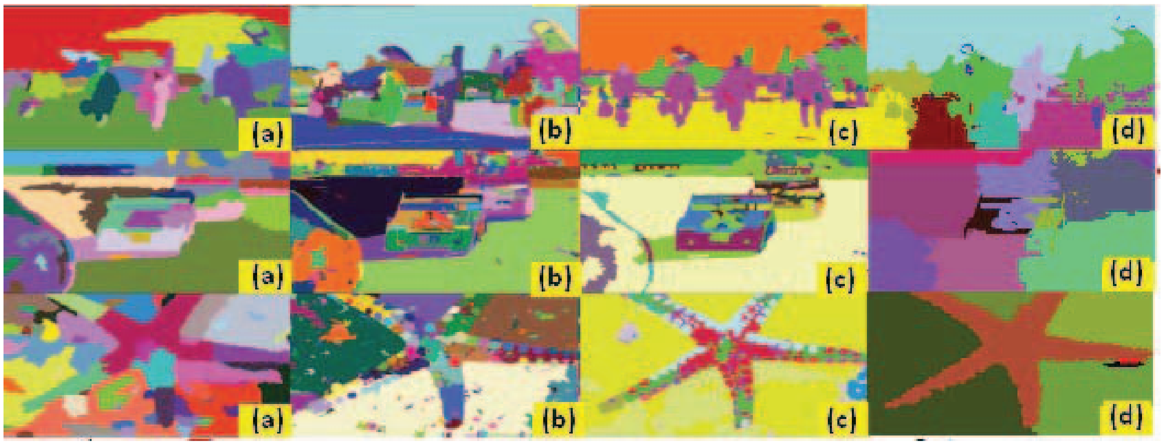

Figure 7: Segmentation results of (a) [11], (b) [9], (c) [10] and (d) the proposed method.

The efficiency of the algorithm can be observed under testing of high resolution image (2560x1920) performance, as given in Figure 8. The resultant segmentation map is obtained in 65 seconds (19 sec for 10000 super pixel generation), which is actually an efficient result according to graph based methods. For most of the graph based methods, it is impossible to execute on large sized images. However, the proposed method is capable of working on any size of images, by arranging the super pixel number.

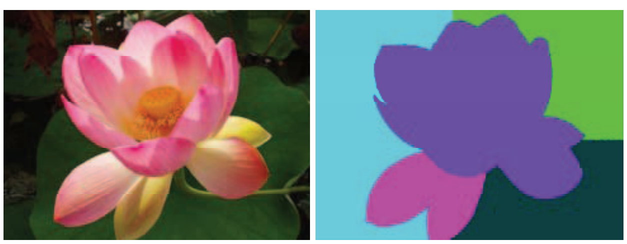

Figure 8: A high resolution image (2560x1920) and the extracted segmentation map.

5. CONCLUSION

In this work, an efficient graph based segmentation algorithm is proposed by exploiting a novel super pixel extraction algorithm. The images are modeled as weighted graphs, whose nodes correspond to super pixels; and normalized cuts are utilized to obtain final segmentation. The constraints on super pixels provide a regular graph increasing the stability of the overall algorithm. According to the results, both super pixel extraction and graph cut segmentation parts are quite efficient compared to the state of the art techniques.

6. ACKNOWLEDGEMET

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